

VisionAssist Smart Glasses System

Complete Line-by-Line & File-by-File Technical Guide

System Architecture Overview

This guide details every single file across the **VisionAssist** smart glasses codebase. The architecture combines React 18 & TypeScript frontend UI, FastAPI & PyTorch Deep Learning backend microservices, YOLOv8 object detection, OpenCV computer vision pipelines, and Supabase serverless edge functions.

1. Backend Microservices (backend/)

- **backend/main.py**

Primary FastAPI Python server. Features PyTorch Currency CNN inference (<50ms), YOLOv8 object detection, OpenCV traffic light color spectrum (HSV) parsing, zebra crossing Hough line transforms, EasyOCR text extraction, Gemini OpenRouter integration, and pyttsx3/Sphinx offline speech engine.

- **backend/train_currency_model.py**

Deep learning PyTorch training script. Defines 4-stage CurrencyCNN, preprocesses dataset images from archive (7), trains model for 20 epochs with Adam optimizer, achieved 83.33% Test Accuracy, and outputs currency_model.pth.

- **backend/currency_model.pth**

Saved PyTorch binary weights checkpoint file for 7-class Indian currency note classification.

- **backend/currency_classes.json**

Metadata mapping 7 neural network class indices to human-readable denomination labels (■100, ■200, ■2000, ■500, ■50, ■10, ■20).

2. Frontend Components (src/components/)

- **src/components/Dashboard.tsx**

Primary Smart Glasses Heads-Up Display (HUD). Features live camera feed canvas, bounding box renderer, navigation HUD, Currency Scanner card with 'Scan Now' button, voice visualizer, emergency SOS trigger, and settings modal.

- **src/components/LandingPage.tsx**

Marketing showcase page highlighting smart glasses features, AI technologies, hardware specs, and system capabilities.

- **src/components/MapPanel.tsx**

Interactive Leaflet GPS map rendering current user position, walking route polyline, destination marker, and map style toggles.

3. Frontend Core Libraries (src/lib/)

- **src/lib/detection.ts**

Client-side vision library. Captures JPEG frames from video, handles detectCurrency() failover, runs local COCO-SSD detection, and renders glowing bounding boxes.

- **src/lib/speech.ts**

Multi-lingual speech engine. Translates announcements and voice commands into English, Tamil, Hindi, Telugu, and Kannada.

- **src/lib/VoiceEngine.ts**

Web Speech Synthesis API wrapper managing voice queue, pitch, rate, and speech event listeners.

- **src/lib/maps.ts**

GPS navigation library calculating walking distances, ETA, step-by-step turn directions, and location geocoding.

- **src/lib/supabase.ts**

Supabase client setup and database TypeScript interface definitions.

- **[src/lib/storage.ts](#)**

LocalStorage helper utility for persistent application state and offline caching.

4. App Core & Root Files (src/ & root)

- **[src/App.tsx](#)**

Root React application router managing view switching between Landing Page and Dashboard.

- **[src/main.tsx](#)**

Entry point mounting React DOM root into index.html.

- **[src/index.css](#)**

Global Tailwind CSS styles, Inter fonts, glassmorphism utilities, and keyframe animations.

- **[supabase/functions/vision-analyze/index.ts](#)**

Serverless Deno Edge Function proxying Gemini 2.5 Flash Vision requests.

- **[supabase/migrations/*.sql](#)**

PostgreSQL database schemas for detection logs, emergency contacts, and config.

- **[generate_pdf_report.py](#)**

Python ReportLab script programmatically creating PDF documentation.

- **[package.json & vite.config.ts](#)**

NPM package manifest and Vite build tool configuration.